Hour-by-Hour Work Plan

How Stable Are Regression Conclusions Across Subpopulations?
A Finite-Sample Empirical Study

Project execution schedule

July 24, 2026

1 Overview

This schedule expands the original day-by-day plan into an hour-by-hour workflow. It assumes approximately **8 focused hours per day** and keeps the same 12-day target for a first full draft, with optional buffer days afterward.

2 Working Rhythm

Each day is organized as:

- **Hours 1–2:** reading, planning, and note consolidation,
- **Hours 3–6:** coding, computation, experiments, and debugging,
- **Hours 7–8:** writing, documentation, and integration into the paper.

The core rule remains: **write something every day**.

3 12-Day Hour-by-Hour Plan

Day 1: Lock the Scope and the Exact Contribution

Main goal: finalize the question, paper scope, and contribution.
Hour 1: Write the exact research question in one sentence.
Hour 2: Write a short statement of the paper's contribution.
Hour 3: Decide on the 4 simulation scenarios.
Hour 4: Decide on the 4 regression models to compare.
Hour 5: Decide on the 4 main stability metrics.
Hour 6: Set up the project folders and confirm the paper directory structure.
Hour 7: Create the LaTeX paper file and section headings.
Hour 8: Draft a one-page project memo with scope, models, metrics, and scenarios.
Deliverables by end of day:

- 1-page project memo,
- LaTeX skeleton with sections,
- clear list of models, metrics, and scenarios.

Day 2: Literature Review and Introduction Notes

Main goal: complete the small literature review and map the introduction.

Hour 1: Select the 6–10 most relevant papers.

Hour 2: Read and annotate the first half of the papers.

Hour 3: Read and annotate the remaining papers.

Hour 4: Extract notes on subgroup heterogeneity and interaction modeling.

Hour 5: Extract notes on transportability, distribution shift, and prediction versus interpretation.

Hour 6: Write the rough introduction outline: motivation, prior work, gap, contributions.

Hour 7: Start the bibliography file with the key references.

Hour 8: Turn notes into a short introduction draft skeleton.

Deliverables by end of day:

- annotated notes for 6–10 references,
- rough introduction draft,
- `references.bib` started.

Day 3: Build the Simulation Engine

Main goal: get one complete simulation pipeline working from start to finish.

Hour 1: Write the data-generation function for pooled data and subgroup-structured data.

Hour 2: Write the model-fitting functions for pooled and subgroup-aware regressions.

Hour 3: Write helper functions to extract coefficients, signs, and standard errors.

Hour 4: Add RMSE and MAE calculation helpers for within-group and transported prediction.

Hour 5: Wire the functions into one end-to-end simulation script.

Hour 6: Run one pooled-baseline scenario and debug any failing steps.

Hour 7: Save outputs in a clean and reproducible format.

Hour 8: Document the simulation workflow and file naming conventions.

Deliverables by end of day:

- working simulation script,
- one successful end-to-end run,
- clear file-saving workflow.

Day 4: Implement All Simulation Scenarios

Main goal: finish the simulation framework for all scenarios.

Hour 1: Implement the no-heterogeneity pooled-baseline scenario.

Hour 2: Implement the covariate-shift-only scenario.

Hour 3: Implement the coefficient-heterogeneity-only scenario.

Hour 4: Implement the combined-heterogeneity scenario.

Hour 5: Add variation in sample size and collinearity.

Hour 6: Add variation in subgroup balance and run pilot combinations.

Hour 7: Check for bugs, degenerate fits, and unstable outputs.

Hour 8: Clean the code and confirm the full scenario grid runs.

Deliverables by end of day:

- stable simulation code for all scenarios,
- pilot outputs confirming the setup works.

Day 5: Run the Full Simulation Study

Main goal: generate the main simulation results.
Hour 1: Launch the full simulation grid.
Hour 2: Monitor the first batch and confirm outputs are being saved.
Hour 3: Verify replicate-level results and restart if a parameterization fails.
Hour 4: Continue running the remaining batches.
Hour 5: Aggregate preliminary outputs into summary tables.
Hour 6: Inspect sign instability, standardized coefficient drift, transported prediction gaps, and covariate shift summaries.
Hour 7: Save the summary outputs in the results folder.
Hour 8: Write down any anomalies or follow-up checks for the next day.
Deliverables by end of day:

- full raw simulation outputs,
- first summary tables.

Day 6: Create Simulation Figures and Interpret Results

Main goal: turn the simulation output into interpretable figures and findings.
Hour 1: Decide which 4–6 plots best summarize the main patterns.
Hour 2: Build the sign-stability figure.
Hour 3: Build the coefficient-drift figure.
Hour 4: Build the transported-prediction-gap figure.
Hour 5: Build any pooled-versus-subgroup comparison plots.
Hour 6: Choose the best versions and remove weak or redundant visuals.
Hour 7: Write bullet-point interpretations under each figure.
Hour 8: Identify the strongest results that deserve emphasis in the paper.
Deliverables by end of day:

- final versions of core simulation figures,
- notes on the main empirical findings.

Day 7: Select and Prepare Real Datasets

Main goal: choose two real datasets and prepare them for analysis.
Hour 1: Identify candidate datasets with a continuous response and clear subgroup variable.
Hour 2: Select the two datasets that best fit the paper's scope.
Hour 3: Clean and preprocess the first dataset.
Hour 4: Clean and preprocess the second dataset.
Hour 5: Define subgroup variables carefully and check their coding.
Hour 6: Create exploratory tables and plots.
Hour 7: Check sample sizes within each subgroup and note imbalances.
Hour 8: Package the datasets into analysis-ready files.
Deliverables by end of day:

- two analysis-ready datasets,
- initial exploratory summaries.

Day 8: Run the Real-Data Analyses

Main goal: complete the real-data modeling and comparisons.
Hour 1: Fit the pooled model first on the first dataset.
Hour 2: Fit the interaction-based model to the first dataset.
Hour 2: Fit the subgroup-aware models on the first dataset.
Hour 3: Repeat the pooled-first fitting sequence for the second dataset.
Hour 4: Compare pooled coefficients with subgroup-specific coefficients.
Hour 5: Compare within-group and out-of-group prediction performance.
Hour 6: Build 2–4 real-data tables or figures.
Hour 7: Write down how the findings support or contrast with the simulation results.
Hour 8: Clean the outputs and prepare them for insertion into the manuscript.
Deliverables by end of day:

- completed real-data analyses,
- draft figures/tables for Section 4.

Day 9: Write the Methods Section

Main goal: fully draft the methods section.
Hour 1: Write the notation and problem setup.
Hour 2: Define the pooled, interaction, and subgroup-specific models.
Hour 3: Define the stability metrics mathematically.
Hour 4: Describe the simulation design.
Hour 5: Describe the real-data analysis workflow.
Hour 6: Check that notation is consistent across the section.
Hour 7: Insert equations and write transitions between subsections.
Hour 8: Read through the section once and tighten the prose.
Deliverables by end of day:

- full draft of the Methods section.

Day 10: Write the Simulation Results Section

Main goal: write the strongest section of the paper.
Hour 1: Insert the simulation figures and tables into LaTeX.
Hour 2: Draft the narrative for the no-heterogeneity and covariate-shift scenarios.
Hour 3: Draft the narrative for the coefficient-heterogeneity scenario.
Hour 4: Draft the narrative for the combined-heterogeneity scenario.
Hour 5: Highlight when pooled models fail and when interaction models help.
Hour 6: Emphasize the difference between predictive stability and interpretive stability.
Hour 7: Tighten the section so the interpretation stays tied to the research questions.
Hour 8: Check cross-references and figure captions.
Deliverables by end of day:

- full draft of the Simulation Results section.

Day 11: Write the Introduction and Real-Data Section

Main goal: draft the remaining core sections.
Hour 1: Write the motivation paragraph for the Introduction.
Hour 2: Write the literature review and gap statement.
Hour 3: Write the contributions paragraph.
Hour 4: Draft the real-data section structure and dataset descriptions.
Hour 5: Insert the real-data figures and tables.
Hour 6: Explain how the real-data findings connect to the simulations.
Hour 7: Check that the introduction promises only what the paper actually delivers.
Hour 8: Smooth transitions between the Introduction and Real-Data sections.
Deliverables by end of day:

- draft Introduction,
- draft Real-Data section.

Day 12: Write Discussion, Conclusion, Abstract, and Polish

Main goal: complete the first full draft.
Hour 1: Write the Discussion on the main findings.
Hour 2: Add practical implications and limitations.
Hour 3: Add future work and boundary conditions.
Hour 4: Write the Conclusion.
Hour 5: Write the Abstract last.
Hour 6: Clean the references and check citation consistency.
Hour 7: Improve figure captions and notation consistency.
Hour 8: Proofread the full document once from beginning to end.
Deliverables by end of day:

- complete first full draft of the paper.

4 Optional Buffer Days

Day 13: Technical Revision

- Re-run any unstable analyses.
- Simplify any overloaded sections.
- Remove weak figures or unnecessary tables.
- Tighten the claims to match the evidence.

Day 14: Writing Revision

- Improve clarity and flow.
- Shorten repetitive paragraphs.
- Make the introduction sharper.
- Make sure the conclusion is specific and not generic.

5 Checklist of Key Deliverables

By the end of the schedule, you should have:

(1) a working bibliography,

(2) a complete LaTeX manuscript,

(3) a simulation pipeline in R,

(4) two real-data analyses,

(5) 4–6 high-quality figures,

(6) 2–4 summary tables,

(7) one full draft ready for feedback.

6 Priority Order if You Fall Behind

If you fall behind schedule, prioritize in this order:

(1) finish the simulation study,

(2) complete the methods and simulation results writing,

(3) include only two real datasets,

(4) reduce the number of figures,

(5) postpone optional extensions.

7 Final Advice

The paper will be strongest if it remains narrow, empirical, and clear. The simulation study should carry the paper, while the real-data examples should support and illustrate the main message. The goal is not to solve all of subgroup heterogeneity or all of transportability. The goal is to produce a clean, credible empirical paper with a precise question and strong execution.